

Rayhan Nazeel

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EDUCATION

University of Alberta

09/2023 – Present

Bachelor of Science in Computing Science

- Expected Graduation: 2027

PROJECTS

Android Event Lottery Platform (Andriod Studio, Firebase, Java)

09/2025 – 12/2025

- Led development of an **Android event lottery platform** enabling fair access to high-demand community events through waiting lists, randomized selection, QR-based discovery, and real-time notifications.
- Coordinated a **6-member team** using GitHub with structured branching, pull requests, code reviews, and merge workflows.
- Delivered **30+ user stories using an Agile, Kanban-based workflow** tracked in **GitHub Projects**, implementing lottery sampling, replacement draws, role-based access, and **real-time Firebase** updates.
- Integrated **Google Maps API** for event location display and **optional geolocation-based verification**, and oversaw use of UML, CRC cards, and JUnit / Android instrumentation tests to improve reliability.

Ray Tracing Renderer (C)

09/2024 – 12/2024

- Developed a **3D renderer from scratch** generating **realistic images of spheres in 3D space**, achieving **>90% shading accuracy** across test cases.
- Implemented **ray-object intersection, shading, and color computation algorithms** using **dynamic memory** and **low-level file I/O**, optimizing performance for scenes with **50+ objects**.
- Optimized the **rendering pipeline** with **qSort and memory management**, reducing **computation time by ~20%** while simulating **realistic lighting and reflections**.

A* Pathfinding Visualizer (RISC-V, GLIR)

01/2025 – 04/2025

- **Implemented** the A* search algorithm **entirely in RISC-V assembly** to compute optimal paths on a 2D grid.
- **Visualized pathfinding in real time** using the course-provided **GLIR graphics interface**.
- Optimized **heap-based open-list** and memory access patterns for efficient traversal and expansion.
- **Debugged and validated** the algorithm against multiple map test cases using terminal visualization tools.

SKILLS

Programming languages: Python, Java, C, RISC-V Assembly, SQL

Software Development: Data Structures & Algorithms, Object-Oriented Programming

Data Science & ML: NumPy, Pandas, Matplotlib, Reinforcement Learning, Neural Networks

Tools and Frameworks: Git, GitHub, Linux, VS Code, Android Studio, Firebase

LEADERSHIP AND AWARDS

Programming Club, Vice President

06/2022

- Led training sessions mentoring **junior members** on **WRO competition rules, problem-solving strategies, and programming fundamentals**.
- Coordinated **weekly leadership and project meetings**, driving event planning, team alignment, and long-term growth of the club.

Awards & Recognition

05/2023

- 2nd Place, Al Tasamuh National Inter-School Robotics Competition
- University of Alberta Regional Excellence Scholarship – \$5,000
- University of Alberta International Admission Scholarship – \$5,000